Music Piece Classification

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STAT 4830

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MLP results in summary

Unperturbed (no overlap?)

- 2487 train, 521 test snippets
- **0.344 top-1 accuracy, 0.564 top-5 accuracy** (1 hidden layer MLP [512])

Previous perturbed (63820 train, 13250 test snippets) - data leakage

- tempos [0.9, 0.95, 1.0, 1.05, 1.1], pitches [-2, -1, 0, 1, 2]
- 8s snippets, 4s overlap (i.e. '4s hop')
- **0.8775 top-1 accuracy, 0.9655 top-5 accuracy**

Revised perturbed v1 (41850 train, 10925 test snippets) - fixed data leakage

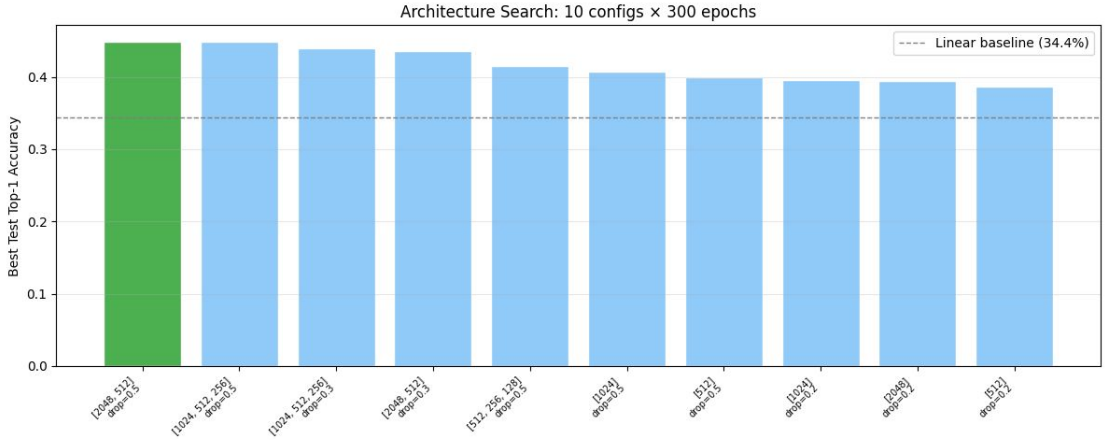
- tempos [0.8, 0.9, 1.0, 1.1, 1.2], pitches [-2, -1, 0, 1, 2]
- 8s snippets, 2s overlap (i.e '6s hop')
- **0.4422 top-1 accuracy, 0.6767 top-5 accuracy**

Revised perturbed v2 (75330 train, 19665 test snippets) - v2 of above

- tempos [0.8, 0.9, 1.0, 1.1, 1.2], pitches [<u>-4, -3</u>, -2, -1, 0, 1, 2, <u>3, 4</u>]
- 8s snippets, 2s overlap (i.e '6s hop')
- **0.4287 top-1 accuracy, 0.6608 top-5 accuracy**

MLP Search on Revised Perturbed Dataset v1

	hidden_dims	dropout_p	n_params	best_test_top1	best_test_top5	final_test_top1	final_lr
1	[2048, 512]	0.5	2,836,885	0.4491	0.6781	0.4457	1.25e-04
2	[1024, 512, 256]	0.5	1,551,253	0.4483	0.6930	0.4445	2.50e-04
3	[1024, 512, 256]	0.3	1,551,253	0.4390	0.6778	0.4363	1.25e-04
4	[2048, 512]	0.3	2,836,885	0.4362	0.6662	0.4340	1.25e-04
5	[512, 256, 128]	0.5	611,989	0.4149	0.6891	0.4056	1.00e-03
6	[1024]	0.5	1,204,629	0.4073	0.6294	0.4046	1.25e-04
7	[512]	0.5	602,517	0.3993	0.6155	0.3959	1.25e-04
8	[1024]	0.2	1,204,629	0.3952	0.6085	0.3928	1.25e-04
9	[2048]	0.2	2,408,853	0.3941	0.6032	0.3829	5.00e-04
10	[512]	0.2	602,517	0.3866	0.6094	0.3866	3.13e-05



Best config: [2048, 512], 0.5 dropout

Note: best configurations here are different than that of the unperturbed data

After retrain w/ 1000 epochs:

Final test top-1: 0.4422

Final test top-5: 0.6767

Finetuning

Goal: To finetune some transformer layers of MERT using our data to adapt how it creates embeddings to support our classification purposes. By doing this training along with a classification head with weights from previous well-performing MLP, it provides a better starting point.

See next slide for the 2 phases

Finetuning

- **Phase 1:** Train MLP classifier on precomputed embeddings
  - 768 → BatchNorm → 512 → ReLU → Dropout(0.3) → 430
- **Phase 2:** Unfreeze top n layers of MERT, using weights from Phase 1 as starting point, train end-to-end with audio WAV files (i.e. train those MERT layers in combination with following MLP)
  - Have tried top 2 and top 4 layers
  - Train for 30 epochs - save .pt and log

Finetuning Results - Perturbed Dataset

2 layers unfrozen

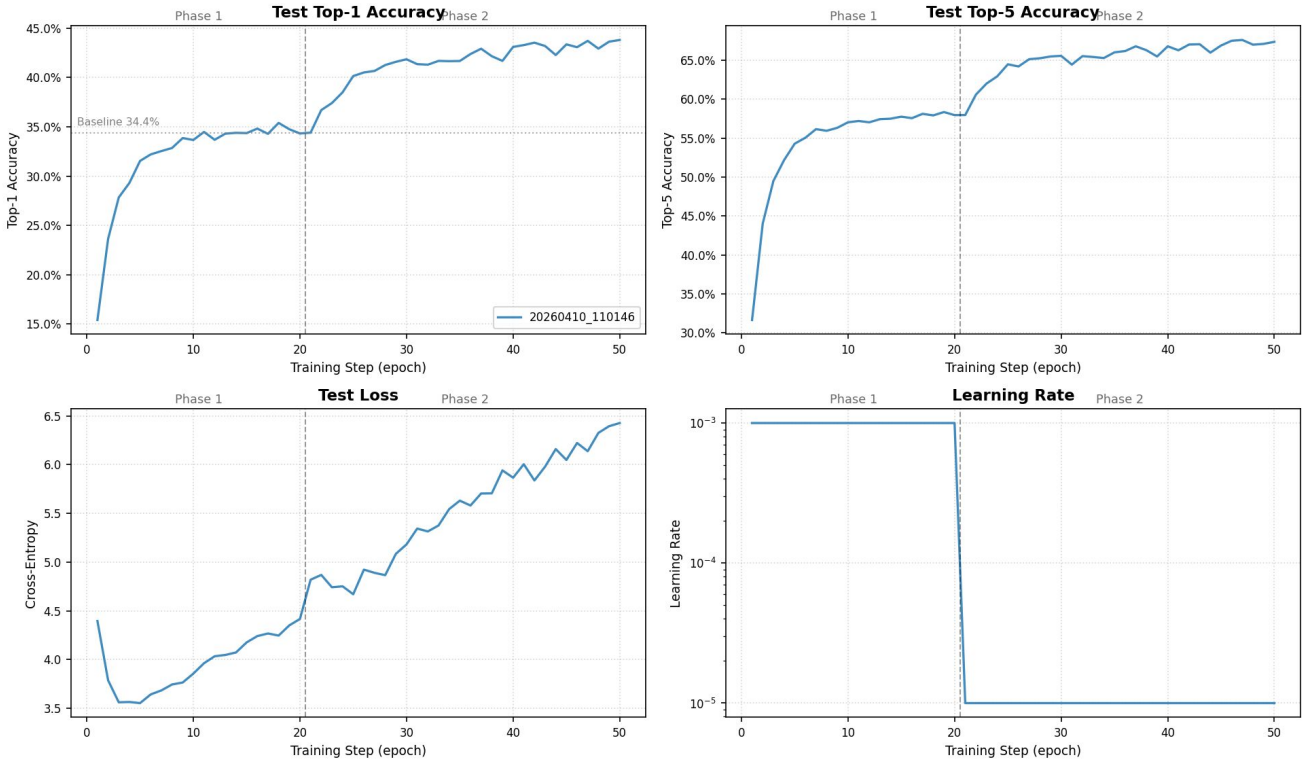
<u>Accuracies</u>

final top-1: 43.8%

best top-1: 43.8%

final top-5: 67.4%

best top-5: 67.6%



Finetuning Results - Perturbed Dataset

4 layers unfrozen

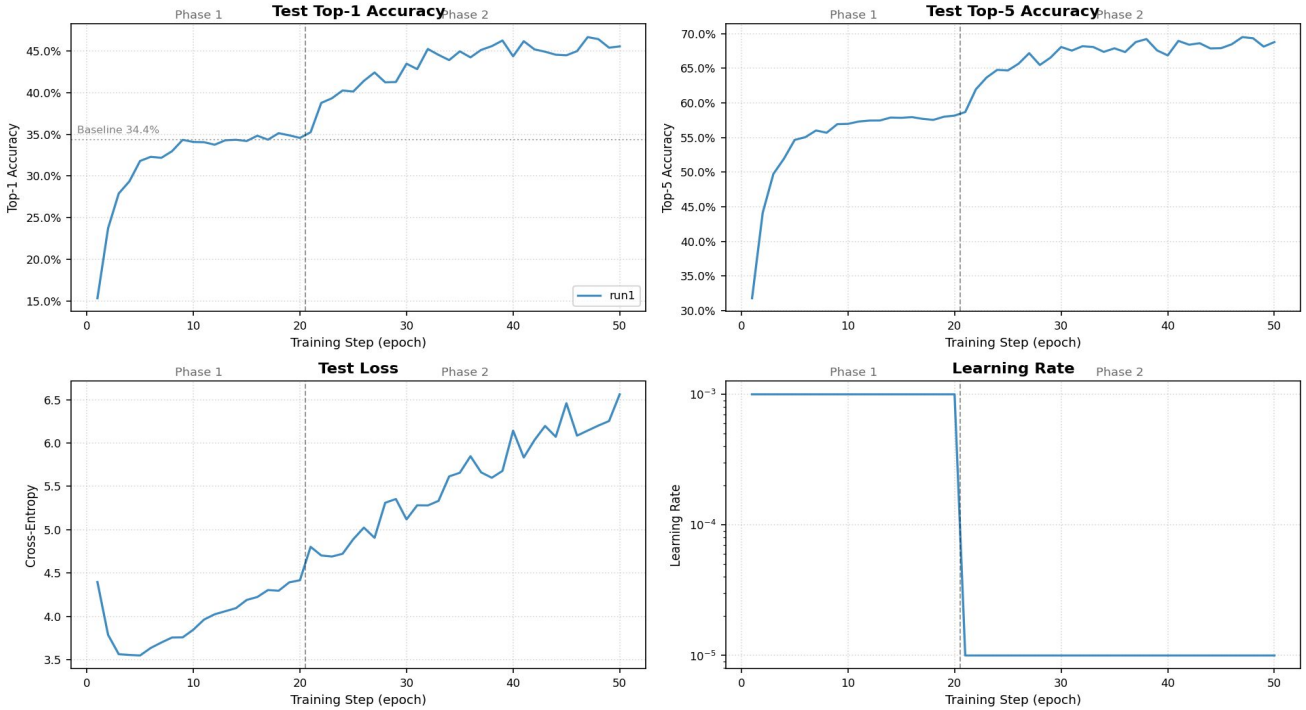
<u>Accuracies</u>

final top-1: 45.6%

best top-1: 46.7%

final top-5: 68.8%

best top-5: 69.5%



Finetuning Results - Perturbed Dataset

- Results currently are comparable to that of just the regular MLP on perturbed dataset
- Upon further analysis, realized that the MLP architecture used in the finetuning version ([512] + 0.3 dropout) was ideal for unperturbed dataset but one of the worst for the perturbed dataset
- Perturbed dataset performed better with more layers generally - best was [2048, 512] + 0.5 dropout → will redo with this

Fine-Tuning vs. Perturbation Robustness

Setup

Two fine-tuned MERT models (2-layer unfreeze, same architecture) trained on different data:

- Model A: trained on 2,487 clean snippets
- Model B: trained on 41,850 perturbed snippets (25 tempo/pitch variants)

Experiment 1 — Does perturbation training help robustness?

Evaluate both models on the perturbed test set (10,925 snippets, all 25 variants).

- Model A result: pending
- Model B result: 46.7% top-1 / 69.5% top-5

Fine-Tuning vs. Perturbation Robustness (cont.)

Experiment 2 — Does perturbation training hurt clean accuracy? Evaluate both models on the 437 clean (t1.0p0) test snippets.

- Model A result: 68.9% top-1 / 86.0% top-5 (on full clean test set of 521)
- Model B result: pending

Interpretation

If Model A degrades significantly on perturbed audio → fine-tuning on clean data learns audio-specific features that don't transfer across perturbations.

If Model B underperforms Model A even on clean audio → perturbation training disrupts the fine-tuning signal without compensating with robustness.

Further Work

1. **Error analysis of finetuning results** - do similar analysis as before as to which pieces are correctly/wrongly classified
2. **Tuning optimizers** - follow some ideas discussed in class
3. **Metric learning** - potentially analysis of classification results olf this too
4. **Scalable experimentation** - hyperparameter sweeps on Prime Intellect